FORECASTING OF PERSISTENT COASTAL WARMING EVENTS IN PERU NOT ASSOCIATED WITH ENSO USING MACHINE LEARNING APPROACHES

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1. Statement and Justification of the Topic

In recent decades, the Peruvian coast has experienced marine heatwave events (Hobday et al., 2016) that are not associated with the ENSO cycle (El Niño–Southern Oscillation). These events may vary in duration and intensity, ranging from relatively short episodes (5–29 days) to prolonged and persistent events (≥30 days). A notable case was the 2017 marine heatwave, which extended from January to April (90–120 days), producing extreme rainfall, impacts on infrastructure, and alterations in the marine ecosystem, and occurred during a neutral ENSO period (Echevin et al., 2018; Garreaud, 2018; Peng et al., 2018). This type of event, which does not directly depend on ENSO evolution, is poorly predictable and reveals a critical limitation in current climate forecasting systems.

The central problem lies in the lack of forecasting models capable of capturing the nonlinear, multiscale, and regional complexity that characterizes atmosphere–ocean processes along the coast of Peru. General circulation models (GCMs), although effective in simulating ENSO at large scales, lack resolution and present biases that limit their ability to adequately represent coastal processes and, in particular, events such as those of 2017. Specifically, they struggle to reproduce events whose development is conditioned by the interaction of local, regional, and remote forcings: coastal upwelling forced locally by coastal winds, the passage of equatorial Kelvin waves, and the South Pacific anticyclone (Echevin et al., 2018; Quispe-Ccalluari et al., 2018).

In light of these limitations, deep learning models emerge as a promising alternative. In particular, convolutional neural networks (CNNs) have demonstrated their ability to overcome some of the limitations of GCMs and to produce more accurate long-term predictions (Ham et al., 2019). Furthermore, these models offer the possibility of formulating and testing hypotheses, thereby contributing to a better understanding of the dynamics of climate events (Rivera et al., 2023). Their strength lies in the ability to identify complex spatial and temporal patterns in fields such as sea surface temperature, winds, and ocean heat content—patterns that remain invisible to traditional

techniques such as Linear Inverse Modelling (LIM) (Jebri and Khodri, 2023) or Reservoir Computing (RC) (Nadiga, 2021).

The justification for this project rests on three dimensions. First, from a scientific perspective, the study will contribute to a better understanding of the dynamics of marine heatwaves from a nonlinear and multiscale perspective, providing evidence on regional mechanisms that operate independently of the ENSO cycle. Second, from a methodological perspective, it will allow for the introduction and validation of CNNs in regional climate forecasting, comparing their performance against classical statistical approaches. Finally, from an applied perspective, the project aims to provide a more robust forecasting tool that can be integrated into early warning systems at scientific and environmental management institutions such as IMARPE, thereby reducing socioeconomic vulnerability to extreme coastal events.

In summary, the development of forecasting models based on CNNs, LIM, or RC represents an opportunity to advance both scientific knowledge and the practical management of climate risk in Peru, in a context where the frequency and intensity of extreme events are exacerbated by climate change.

2. Thesis Objectives

General Objective

To develop and evaluate forecasting models for persistent marine heatwave events in Peru not associated with the ENSO cycle, based on convolutional neural networks (CNNs), linear inverse models (LIM), and reservoir computing (RC).

Specific Objectives

1. To characterize marine heatwave events similar to those observed in 2017 off the Peruvian coast, occurred in recent centuries, differentiating those associated with and independent from the ENSO cycle.
2. To design and implement models based on CNNs, LIM, and RC that leverage the spatial and temporal structure of oceanic and atmospheric fields to capture patterns linked to coastal events.
3. To compare the forecasting performance of CNNs, LIM, and RC in terms of accuracy, stability, and sensitivity across different prediction horizons and data availability scenarios.
4. To assess the operational applicability of the best-performing model for integration into climate early warning systems, with the aim of strengthening risk management in the face of extreme oceanic events.

3. Methodology and Work Plan

To achieve the objectives of this research, the following methodology is proposed:

1. Literature Review
   - Review of previous research on Peruvian coastal warming and its relationship with regional atmospheric-oceanic processes.
   - Review of statistical models used in climate forecasting (CNN, LIM, RC, ensemble models, etc.) and their applications in oceanography and climatology.

2. Database Construction
   - Integration of oceanographic and atmospheric data, both observational and reanalysis (Mercator, ERA5), as well as simulated (CMIP6).
   - Identification of coastal warming events associated and not associated with ENSO.
   - Data preprocessing: interpolation, spatial homogenization, variable normalization, and anomaly calculation.

3. Model Design and Implementation
   - Implementation of convolutional neural networks (CNNs), linear inverse models (LIM), and reservoir computing (RC), adapted to the regional scale of the southeastern Pacific.
   - Model training using transfer learning: pre-training with historical simulations and final adjustment with observational and reanalysis data.
   - Definition of model evaluation metrics: correlation, RMSE, probability of detecting extreme events, and predictive stability across different time horizons.

4. Model Comparison
   - Comparison of the forecasting performance of CNNs, LIM, and RC under scenarios of different data availability (long vs. short windows) and different forecast horizons (1 to 12 months).

5. Operational Evaluation and Validation
   - Sensitivity tests and cross-validation to estimate the robustness of the best model.
   - Assessment of the potential of the best-performing model for integration into an operational climate early warning system along the Peruvian coast.

6. Thesis Writing

In Table 1, an activity schedule is presented:

	2025			2026											
Activity	Oct.	Nov.	Dec.	Jan.	Feb.	Mar.	Apr.	May.	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.
Literature review	*	*	*	*											
Data acquisition	*	*	*	*	*										
Design and implementation				*	*	*	*	*							
Comparison							*	*	*	*	*				
Operational evaluation and validation										*	*	*	*		
Thesis writing								*	*	*	*	*	*	*	*

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